

Roktim Mascharak

PhD Candidate, London School of Geometry and Number Theory (UCL, KCL, Imperial)

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RESEARCH INTERESTS

Algebraic geometry and birational geometry, with a focus on the **Minimal Model Program** (MMP) and the **birational geometry of holomorphic foliations**.

EDUCATION

October 2023–present	PhD in Mathematics, London School of Geometry and Number Theory (UCL, KCL, Imperial). Supervisor: Professor Calum Spicer; co-supervisor: Professor Paolo Cascini.
August 2020–July 2023	MSc in Mathematics (First Class, with Distinction), Tata Institute of Fundamental Research (TIFR), Mumbai. Master's thesis: <i>Existence of Flips and the Minimal Model Program</i> . Thesis supervisor: Professor Omprokash Das. Total CGPA: 7.7.
August 2017–March 2020	BSc (Honours) in Mathematics, Presidency University , Kolkata. Total CGPA: 8.40

AWARDS AND HONOURS

2023–present	LSGNT Doctoral Studentship (joint PhD programme of University College London, King's College London, and Imperial College London).
2020–2023	Institute Fellowship, Tata Institute of Fundamental Research, Mumbai.
2020	All-India Rank 128, IIT-JAM (Mathematics).

PUBLICATIONS AND PRE-PRINTS

1. *Foliated Minimal Models and Flops*. Joint work with Paolo Cascini and Calum Spicer. Pre-print. <https://arxiv.org/abs/2608.09663>
2. *On the Log Sarkisov Program for Foliations on Projective 3-Folds*. Published, **Math. Z.** 312, 124 (2026). <https://doi.org/10.1007/s00209-026-04001-y>, <https://arxiv.org/abs/2406.09434>.
3. *ACC for log canonical thresholds for algebraically integrable foliations*. Joint work with Omprokash Das and Jihao Liu. Published, **Sel. Math.** New Ser. 32, 70 (2026). <https://doi.org/10.1007/s00029-026-01179-1>.
4. *Log Canonical Minimal Model Program for Co-rank One Foliations on Threefolds*. Joint work with Priyankur Chaudhuri <https://arxiv.org/abs/2410.05178>.
5. *On the Log Canonical MMP for Kähler 3-Folds*. Pre-print. <https://arxiv.org/abs/2306.07539>.

FORMALIZATION PROJECTS

1. *Formalizing the statement of Hodge Conjecture*. Part of a group project initiated at the *Formal Conjectures* workshop. <https://github.com/Paul-Lez/HodgeConjecture>.

INVITED TALKS AND SEMINARS

August 2026	Uniqueness of rank one foliated minimal models in arbitrary dimension, Foliations, Fibrations and Moduli, Westlake University, China.
June 2026	<i>Invitation to the Foliated MMP for log canonical foliations</i> , CALF Seminar, Nottingham.
May 2026	<i>Introduction to the Foliated MMP for log canonical foliations</i> , Birational Geometry Seminar, UCLA (online).
September 2025	<i>Invitation to the Foliated MMP</i> , New Trends in Moduli Inspired by K-Stability, Nottingham.
November 2024	<i>What is MMP?</i> , Junior Geometry Seminar, Imperial College London.
September 2024	<i>Let's Blow-up</i> , QED Seminar, Presidency University, Kolkata.
June 2024	<i>The Sarkisov Program for Foliations</i> , LSGNT first-year talk, University College London.
November 2022	<i>Inversion of Adjunction</i> (Algebraic Geometry pre-print seminar), TIFR Mumbai.
June 2022	<i>Existence of Flips and the Minimal Model Program</i> (Master's thesis presentation), TIFR Mumbai.
April 2021	<i>The Thom Isomorphism</i> , TIFR Mumbai.

POSTERS

November 2025	Algebraic Geometry Northeastern Series (AGNES), Amherst.
October 2025	Moduli Problems in Birational Geometry and Foliation Theory, Pisa.
July 2025	Summer Research Institute in Algebraic Geometry, Colorado.
March 2025	Winter School in K-stability, CIRM, Marseille (5-minute talk).

CONFERENCES, SCHOOLS AND RESEARCH VISITS

September 2026	Formal Conjectures Workshop, London
July 2026	Junior Algebraic Geometry Workshop, University of California, Los Angeles.
July 2026	Research visit to the University of Nottingham, hosted by Professor Hamid Abban.
May 2026	Simons Collaboration on Moduli of Higher-Dimensional Varieties, Ann Arbor.
April 2026	Equivariant K-stability, Edinburgh.
July 2025	Summer Research Institute in Algebraic Geometry (bootcamp and main conference), Colorado.
August 2024	Birational Geometry and Foliations, Shanghai.
May 2022	Advanced Instructional School on Intersection Theory, IIT Bombay.
December 2018	Mini MTTS Training Camp, Kolkata.

SEMINAR ORGANISATION

Sep–Dec 2024	Organiser, LSGNT Lunch Seminar, KCL, London.
Sep 2022–Jan 2023	Co-organiser (with Priyankur Chaudhuri), BCHM Learning Seminar (Existence of Minimal Models for Varieties of Log General Type), TIFR Mumbai.
Oct–Dec 2022	MMP Seminar (Student Algebraic Geometry Seminar), TIFR Mumbai.

TEACHING EXPERIENCE

2024–present Senior Teaching Assistant / Teaching Assistant, Imperial College London — Analysis 2, Linear Algebra and Groups, and Introduction to University Mathematics (multiple terms).
Jan–Apr 2025 Teaching Assistant, Calculus 2, King’s College London.
May 2023 Teaching Assistant, mini-courses on the Minimal Model Program and Rationality of Algebraic Varieties, Advanced Instructional School on Birational Geometry, Chennai Mathematical Institute.

OUTREACH AND SERVICE

Taught mathematics to school students at the Mumbai Math Circle (organised by TIFR Mumbai). Participated in the TIFR Frontiers of Science outreach programme.

LANGUAGES

Proficiency English (fluent); Hindi; Bengali (native).